Rishav Gupta

Ph.D. Student, Department of Computer Science and Electrical Engineering
Future Sensing and Interaction Lab, ITE 230, University of Maryland, Baltimore County

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Research Interests

Mobile health sensing and mobile systems, with an emphasis on acoustic and ultrasonic sensing, digital signal processing, and machine learning for cuff-free cardiovascular monitoring on commodity smartphones.

Education

University of Maryland, Baltimore County Jan. 2026 – Present
Ph.D. in Computer Science *Baltimore, MD*
- Advisor: Dr. Dong Li, Future Sensing and Interaction (FSI) Lab.

University of Maryland, Baltimore County Jan. 2025 – Dec. 2025
M.S. in Computer Science *Baltimore, MD*
- GPA: 4.00/4.00. Coursework: Design and Analysis of Algorithms, Machine Learning, Neural Networks.

Manipal Institute of Technology Jun. 2015 – May 2019
B.Tech. in Information Technology *Manipal, India*
- Ranked 8th out of 132 students in the graduating class.

Research Experience

University of Maryland, Baltimore County Jun. 2025 – Present
Graduate Research Assistant, Future Sensing and Interaction (FSI) Lab *Baltimore, MD*
- **Smartphone blood-flow sensing** [2, 4]. Recovering arterial blood-flow waveforms and Doppler indices through the skin via phase-based ultrasonic Doppler; first author of the real-time demonstration [2].
- **Acoustic tissue characterization** [1]. First-author work estimating soft-tissue relaxation time from smartphone acoustic chirps and echoes, validated against laboratory indentation (phantom and human forearm).
- **Voice-scam protection** [3]. Contributed to Ultrasound Watermark, which binds live call audio to the caller's articulatory motion sensed with smartphone ultrasound.

Publications

[1] **Rishav Gupta**, Alyssa N. Maguina, Sithumina Weerarathna, Janelle P. Clark, and Dong Li. "WiP: Smartphone-based Acoustic Tissue Characterization for Personalized Body-contact Systems." In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Work-in-Progress Track, pp. 1693–1696. ACM, 2026. [Paper]

[2] **Rishav Gupta**, Shirui Cao, Jie Xiong, Sunghoon Ivan Lee, Jeremy Gummeson, and Dong Li. "Demo: Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones." In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Demo Track, pp. 1777–1779. ACM, 2026. [Paper] [Code]

[3] Renzhi Hao, Shirui Cao, Yizhu Wen, **Rishav Gupta**, Dharani Nadendla, Mehran Kafai, Hanqing Guo, and Dong Li. "Ultrasound Watermark: Real-time Acoustic Watermarking against Voice Scams on Smartphones." In *Proceedings of the 9th ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP '26)*, co-located with ACM/IEEE SEC. IEEE, 2026. To appear.

[4] Shirui Cao, Jie Xiong, **Rishav Gupta**, Sunghoon Ivan Lee, Jeremy Gummeson, and Dong Li. "Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones." In *Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services (MobiSys '26)*, Emerging Ideas Track. ACM, 2026. [Paper] [Code]

Honors and Awards

First Place (Graduate), ACM Student Research Competition, ACM MobiCom 2026 — Oct. 2026
- For *Phonotonos*; advanced to the ACM SRC Grand Finals.

COEIT Computing for Social Good Award, Research Day, UMBC — May 2026
- One of only two awards presented across 114 posters; conferred by Dean Jeanne M. van Briesen.

First Place, Maryland NEW VENTURE™ Program Pitch Night, bwtech@UMBC — May 2026
- Awarded for *PocketCardio*, a smartphone-based cardiovascular screening venture.

Best Poster Award, CSEE Research Day, UMBC — Apr. 2026
Best Poster Award, COEIT Research Day, UMBC — May 2025

Industry Experience

Gojek (GoTo Group) — Feb. 2022 – Sep. 2024
Senior iOS Engineer — *Bengaluru, India*
- Reduced annual platform cost by $3M by replacing a third-party analytics SDK (CleverTap) with an in-house, open-source tool at full feature parity.
- Cut development time by 20% across the engineering team by automating the bi-weekly CI/CD release pipeline in Python and Shell on GitLab.
- Doubled QA throughput with a real-time Event Visualizer for verifying analytics events in situ, and cut roughly 3 MB from the app binary through SwiftSyntax AST-based dead-code detection.

Nuclei — Jul. 2019 – Feb. 2022
Senior iOS Developer and Team Lead — *Bengaluru, India*
- Founding iOS developer of FedMobile (Federal Bank); implemented 60% of shipped features, including Payments, Investments, Loans, Onboarding, and Card Management.
- Promoted to Team Lead; led four iOS developers and closed a four-month delivery gap between the iOS and Android platforms.

Hera doo — Jun. – Jul. 2018
Android Developer Intern — *Mostar, Bosnia*
- Built an end-to-end Android sales-order application from scratch, increasing field sales by 25%.

Service and Leadership

Chair, Graduate Assistant Advisory Committee (GAAC), UMBC Graduate Student Association — 2026 – 2027
Lead Volunteer, ACM/IEEE Symposium on Edge Computing (SEC 2025) — Dec. 2025
HR Head, ISTE Students' Chapter, Manipal Institute of Technology — 2017 – 2018
Co-Event Head (Acumen), TechTatva '17, Manipal Institute of Technology — Oct. 2017
Management Committee Member, ISTE Students' Chapter, Manipal Institute of Technology — 2016 – 2017

Technical Skills

Languages	Swift, Python, Java, Shell, SQL, HTML/CSS
Methods	Acoustic and ultrasonic signal processing, physiological sensing, tissue biomechanics, embedded audio, machine learning
Tools	Xcode, SwiftUI, Android Studio, Git, GitLab CI/CD, LaTeX

References

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